

SHIVANGI TRIPATHI

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SUMMARY

Ph.D. candidate in Computer Science at Texas State University specializing in trustworthy generative AI. Research spans hallucination detection and mitigation across text-based large language models (LLMs), vision-language models (VLMs), and audio-language models—covering adversarial robustness evaluation, paragraph-level misinformation correction, vision-grounded caption faithfulness, and encoder-decoder attribution in clinical audio settings. Work is published at IEEE MobiSecServ, AITest, and TPS, with ongoing studies on multimodal hallucination benchmarking.

EDUCATION

Doctor of Philosophy (Ph.D.) in Computer Science <i>Texas State University, San Marcos, TX</i>	Aug 2023 – Present <i>Expected 2027</i>
• Research focus: hallucination detection, trustworthy AI, LLM/VLM/audio-LLM reliability, machine learning security	
Master of Science (M.S.) in Electrical Engineering <i>Indian Institute of Technology (IIT) Jodhpur, India</i>	Jul 2021 – Jul 2023
• Research areas: cyber-physical systems, 6G networks, ML for wireless communications, mobile edge computing	
Bachelor of Technology (B.Tech.) in Electronics & Communication Engineering <i>Guru Gobind Singh Indraprastha University (GGSIPU), Delhi, India</i>	Aug 2017 – May 2021

TECHNICAL SKILLS

- **Languages:** Python, C++, Java, MATLAB
- **ML & AI:** PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, Keras
- **LLMs & Multimodal:** Transformers, BERT, GPT-style models, LLaVA, CLIP, Qwen2-Audio, RAG, prompt engineering, hallucination detection
- **NLP & Data:** spaCy, NLTK, FAISS, LangChain, Pandas, NumPy, feature engineering
- **Infrastructure:** AWS, GCP, Docker, Linux, Git

RESEARCH EXPERIENCE

Doctoral Instructional Assistant / Researcher <i>Human AI Synergy Lab, Texas State University</i>	Aug 2023 – Present <i>San Marcos, TX</i>
• Conduct research on making large language models, vision-language models, and audio-language models more reliable, with a focus on detecting, analyzing, and reducing hallucinations across modalities.	
• Design and run controlled experiments to study model behavior under different prompts, datasets, and evaluation settings.	
• Develop software tools and standardized evaluation workflows in Python to support lab projects and collaborations.	
Research Assistant <i>Wireless Communications Lab, IIT Jodhpur</i>	Jul 2021 – Jul 2023 <i>Jodhpur, India</i>
• Contributed to research on radio resource management and edge computing for next-generation wireless (6G) systems.	
• Built simulation and analysis pipelines to study latency, reliability, and efficiency trade-offs in mission-critical mobile edge computing scenarios.	

SELECTED PROJECTS

REVISE: Correcting Factual Errors in LLM-Generated Text	<i>Jun 2024 – Present</i>
• Built a paragraph-level system that detects and fixes factual mistakes in LLM-generated text by checking model claims against evidence retrieved from trusted sources.	
• Improved factual correctness to 76.8% on a custom benchmark of 1,400+ passages while keeping meaning intact, with FactCC confidence consistently above 0.97 .	
Stress-Testing LLMs Against Adversarial Prompts	<i>Aug 2023 – May 2024</i>
• Designed adversarial prompts that subtly change wording or topic to study when Vicuna-7B and LLaMA-2-7B start hallucinating or producing unsafe answers.	
• Observed attack success rates of 92.53% for small semantic changes and 88.68% for out-of-distribution prompts, and derived practical guidelines for safer deployment.	

VLM Caption Faithfulness: KG-Guided Caption Compression

Jan 2025 – Present

- Developed a knowledge-graph-guided rate-distortion framework (RGCC) that iteratively compresses verbose VLM-generated captions while preserving semantic fidelity, using T2I reconstruction loops and scene-graph comparison (SPICE F1, RelTR, spaCy triples).
- Empirically identified that shorter compressed captions can yield *lower* T2I distortion than the original verbose output—a novel finding with direct implications for hallucination measurement in VLMs.

Audio-LLM Hallucination Attribution in Clinical Audio

Jan 2025 – Present

- Designed a three-level hallucination detection pipeline (audio denial, topic-level semantic similarity via sentence-transformers, sentence-level NLI via DeBERTa) applied to Qwen2-Audio-7B on PriMock57 clinical consultations.
- Pilot study revealed 75% overall hallucination rate and 80% topic mismatch rate, with inference runs indicating encoder signal is not controlling decoder output—motivating mechanistic interpretability follow-up.

Cancer Detection from Histopathology Images using LLMs

Oct 2025 – Dec 2025

- Built an end-to-end system that predicts cancer from histopathology images and uses a BioBERT-based model to read and summarize pathology-style text for decision support.
- Evaluated on PatchCamelyon (PCam), achieving **94.2%** AUC and **0.91** F1-score, a **12%** relative F1 improvement over a CNN-only baseline.

Question Answering on SQuAD with LSTM, Transformer, and BERT

Sep 2024 – Dec 2024

- Built LSTM, Transformer, and fine-tuned BERT models in **PyTorch** for extractive question answering on the SQuAD dataset.
- Set up reproducible training and evaluation pipelines; achieved **83%** F1, more than **20%** higher than a TF-IDF baseline.

Resource Allocation Algorithms for 6G Wireless Networks

Jun 2022 – Jul 2023

- Developed and simulated uplink/downlink resource allocation strategies for high-mobility 6G scenarios in **MATLAB/Simulink**.
- Achieved a **30%** reduction in bit error rate and a **25%** increase in spectral efficiency compared to baseline allocation schemes.

PUBLICATIONS

- **Tripathi, S.**, H. Griffith, H. Rathore.
“Assessing Hallucination in Large Language Models under Adversarial Attacks.”
Proc. IEEE International Conference on Mobile and Secure Services (MobiSecServ), 2024.
- **Tripathi, S.**, T. Jennifer, H. Griffith, H. Rathore.
“REVISE: A Framework for Paragraph-Level Misinformation Correction in Large Language Models.”
Proc. IEEE International Conference on Artificial Intelligence Testing (AITest), 2025.
- **Tripathi, S.**, H. Rathore.
“Detecting and Correcting Hallucinations in Paragraph-Level Text with Ensemble-Based Evaluation.”
Proc. IEEE International Conference on Trust, Privacy and Security in Intelligent Systems (TPS), 2025.
- **Tripathi, S.**, T. Jennifer, H. Griffith, H. Rathore.
“Paragraph-Level Hallucination Detection and Correction for Trustworthy Large Language Models in Networked Systems.”
Proc. IEEE Consumer Communications and Networking Conference (CCNC), 2026.

TEACHING EXPERIENCE

- Guest Lecturer

Department of Computer Science, Texas State University

Spring 2025/2026

San Marcos, TX
- Delivered a lecture on “Security Attacks in LLMs”; communicated technical concepts clearly and facilitated class discussion.
- Teaching Assistant — Computer Security (CS 4318)

Texas State University

Spring 2025

San Marcos, TX
- Assisted in instruction and assignment design; mentored students on secure coding and analytical problem solving.
- Teaching Assistant — Introduction to C++ (CS 1428)

Texas State University

Fall 2023/2024

San Marcos, TX
- Led labs and debugging sessions; reinforced object-oriented programming and foundational software engineering practices.

HONORS AND AWARDS

- Academic Excellence Award, Branch 2nd Rank (top 2 in branch), M.S., IIT Jodhpur, 2023.

RELEVANT COURSEWORK

- **AI/ML:** Machine Learning and Pattern Recognition, Deep Learning, Advanced Data Mining
- **NLP:** Advanced Natural Language Processing
- **Systems/Security:** Secure Cyber-Physical Systems, Network Science, Mobile Networks and Computing

CERTIFICATIONS

- Building AI Agents with Multimodal Models, NVIDIA Deep Learning Institute (DLI), 2025.
 - Developed contrastive pre-training (CLIP-style) pipelines to align RGB and LiDAR embeddings; built cross-modal projectors for cross-modality inference using PyTorch.
 - Explored multimodal fusion strategies (early, late, intermediate) across RGB, LiDAR, audio, and CT scans for building robust AI agents.